INSTITUTIONAL COLLATERAL CONTROL, PROVEN ACROSS CHAINS

An asset stops being good collateral before its price moves.

On-chain credit only has the price axis. This ledger adds the other one — and the destination chain verifies the event itself instead of trusting a monitoring vendor.

LIVE ON CC3 TESTNET

EligibilityLedger 0x00FcAE7CD35919d91322Ba30B42B51b4b379FF32
Real event ingested sNUSD Paused · Ethereum block 25,745,732

60 / 60 local checks
4 forgeries rejected on-chain

TradFi manages collateral on two axes. Chains only carry one.

AXIS	WHAT IT COVERS	ON-CHAIN LENDING
Price	Mark, haircut, volatility	**Oracle, LTV, liquidation**
Eligibility	Redeemability, transferability, issuer control, instrument pause	**Effectively absent**

The ECB defines eligible collateral in Guideline (EU) 2015/510 Part Four and national central banks vet instruments *before* they are usable. ICMA triparty repo maintains eligibility sets by grade; ISDA publishes jurisdictional eligible-collateral tables. Eligibility is a first-class control, not a footnote to price.

THE CORRECTION WE MADE

Our first framing — "the eligibility axis does not exist on-chain" — was **wrong**. Aave can set LTV to zero. Morpho curators can set caps to zero. Hypernative detects and pauses in seconds.

The accurate problem is narrower:

> When an asset-level event fires, the parties holding that asset are **unrelated third-party protocols**. Their only paths are subscribing to the same vendor or acting by hand. No path lets the destination contract **verify the event itself** — and when asset and credit market sit on different chains, the gap widens.

Sources are cited in full in `CROSS_CHAIN_COLLATERAL_ELIGIBILITY_PROJECT_STATE.md §6, §12`. The ECB/ICMA/ISDA framing is documented; we did not obtain a verbatim clause stating eligibility is lost for non-price reasons, and do not claim one.

A real instrument was paused on Ethereum *nine days ago*.

THE SOURCE TRANSACTION WE INGEST

```
asset   sNUSD (Staked NUSD, Neutrl)
        0x08efcc2f3e61185d0ea7f8830b3fec9bfa2ee313
event   Paused(address)
block   25,745,732 · 2026-08-13 11:14:59 UTC
status  receiptStatus = 1
```

Neutrl halted NUSD issuance and redemption. Downstream, **Strata paused its own products separately and by its own decision** — nothing propagated. That silence between an issuer's event and everyone holding the instrument is the entire problem.

HOLDER OF SNUSD AT THAT BLOCK	BALANCE
Pendle Standardized Yield SY-sNUSD	8,202,400
Morpho Blue singleton	1,828,755
Uniswap V4 PoolManager	7,710

Morpho's sNUSD/USDC market still carried **$1,661,397 borrowed** against a reported collateral value of **$0**.

STATED LIMITATION

This incident was **not cross-chain** — issuer, wrapper and lender were all on Ethereum. It proves the event class is real. The cross-chain topology is a separate claim with separate evidence, on the next slide. We do not merge the two.

Nine-figure collateral is already lent on a chain that cannot see the issuer.

USDE → ROBINHOOD CHAIN

Controlled on Ethereum. Bridged by LayerZero OFT. Lent against on Morpho.

$285,532,168
collateral posted

$251,647,632
borrowed · 92% LLTV

SYRUPUSDG → ROBINHOOD CHAIN

Maple's pool stays on Ethereum; the collateral market does not.

$90,801,145
collateral posted

$79,438,999
borrowed

CENTRIFUGE JAAA / JTRSY

The split is encoded in the protocol itself: `PoolId >> 48` names the hub chain. Both resolve to **hub = Ethereum**.

spoke Avalanche	250,000,001
spoke Base	48,207,418

A Base collateral market is already open.

All figures read directly from chain state and the Morpho public API, 2026-08-18. **The honest counter-argument:** on a single chain you would just read the issuer contract, and that is simpler. This project's scope is precisely the case where you cannot.

The ledger has no oracle and no owner-write path. A proof is the only way in.

SOURCE · ETHEREUM

sNUSD.Paused()

block 25,745,732
an ordinary mainnet transaction

—proof→

ATTESTCOIN

Merkle inclusion + block continuity

Proof Builder API
siblings 10 · continuity roots 69

→

CREDITCOIN CC3

EligibilityLedger .submitEvent()

BlockProver precompile 0x…0FD2
then five checks, in order

1	2	3	4	5
Replay	**Inclusion**	**Success**	**Signature**	**Emitter**
keyed by `keccak(chainKey, height, txIndex)` — position, recomputed on-chain	the precompile. A bad proof reverts inside the runtime	`receiptStatus` == 1. The precompile does not check this — we do	is this a configured impairment or restoration event?	allowlist. Without it, anyone deploys a look-alike and proves it

NO_PROOF

▶ PROOF ADMITTED

IMPAIRED

GatedCreditLine.openPosition() → revert AssetImpaired.

Checks 4–5 *skip* non-matching logs rather than reverting — real source transactions are batched; one we ingested carried 48 logs.

Attestcoin is **inbound-only**: CC3 can verify foreign transactions but cannot write back to them. So the credit venue itself must live on CC3. That constraint shaped the architecture rather than being worked around.

It runs. One command deploys it and ingests the real event end to end.

CONTRACT	ADDRESS ON CC3 TESTNET
EligibilityLedger	`0x00FcAE7CD35919d91322Ba30B42B51b4b379FF32`
GatedCreditLine	`0xDcd5dfb82CA8fcDD3E628b327d5CeaCdBE69771A`
EvmV1Decoder (our build)	`0x00718d01897236C3352D5BA689BcC5c540dd9d97`
BlockProver precompile	`0x0000000000000000000000000000000000000FD2`
Admitting transaction	`0xeca0db64…1ff2461c` · 346,598 gas

Re-verified on-chain in `docs/transcripts/verify-cc3-2026-09-07.log`.

We deploy our own decoder because the one listed on the official chains & environments page is an older build missing `getLogsByEventSignature`.

LOCAL REGRESSION

eligibility.test.js	50 / 50
realdata-eligibility.test.js	10 / 10
ledger.test.js (v1 regression)	15 / 15
realdata.test.js	8 / 8
Total	83 / 83

The real-data suites fetch **live Attestcoin proofs for real mainnet transactions** and run them through the real decoder. Only the BlockProver precompile is mocked locally — it is a Creditcoin runtime component that cannot exist on a local EVM, and the deployed scripts exercise the real one.

Four forgeries, four different refusals, on-chain — and the ledger never moved.

ATTEMPT	REFUSED BY	REVERT
A Replay the proof already admitted	LEDGER	`QueryAlreadyProcessed`
B A genuine proof of a genuine mainnet transaction — emitter not allowlisted	LEDGER	`NoMatchingEvent`
C The same bundle, Merkle root flipped by **one bit**	RUNTIME	`"Merkle proof validation failed"`
D The real mainnet proof, re-labelled as Sepolia	RUNTIME	`"Continuity proof does not match attestation or checkpoint"`

C and D never reach our code. The BlockProver precompile refuses them inside the Creditcoin runtime — which is the point of building on Attestcoin rather than re-implementing verification.

The whole demo runs as `eth_call` against the live contracts, so it cannot write, and it reads the ledger back afterwards to show `IMPAIRED · impairedSince 25745732 · events 1` is unchanged.

TWO RULES WORTH ARGUING ABOUT

— **Gate new risk, never trap users.** `repay` and `withdrawCollateral` stay open forever. Only new exposure is refused. A control that blocks the exit is a hostage situation.

— **Earliest proven impairment wins.** A later proof never pushes the cutoff forward. Credit extended after the real event does not become sound because the proof arrived late.

The prior art is strong. Here is exactly what is left.

NOT CLAIMED

— **Not real-time protection.** Measured attestation lag is 8–9 minutes. This is delayed but verifiable gating of *new* credit. Hypernative binds detection to pre-approved on-chain actions and responds in seconds — on speed we lose.

— **`NO_PROOF` is not a clean bill of health.** Absence is unprovable. It means only that nothing has been proven here yet.

— **Not control over existing markets.** Morpho Blue's core has no pause primitive at all — zero matches for pause/freeze/emergency/blacklist in `Morpho.sol`. We apply eligibility logic to a venue on CC3; we do not reach into other chains.

— **Discovery is not trustless.** Choosing *which* transaction to submit still needs off-chain log scanning. Verification is trustless; discovery is not.

PRIOR ART, NOT UNDERRATED

WHO	WHAT THEY ALREADY DO
Aave	Guardian + Risk Steward paused rsETH across five networks simultaneously, April 2026
Chaos Labs	Edge Risk Oracle injects parameters automatically via Chainlink Automation
Hypernative	Detection bound to on-chain action in seconds — and was Neutrl's own pauser
Chainlink	MVR feeds can technically carry status; no production eligibility feed found

> In every one of these the trigger is a **human or a market-risk metric**, and the destination **trusts a vendor**. What is missing is narrow and specific: an issuer-side event, carried across a chain boundary, that the destination credit contract verifies for itself. That is the primitive this repo ships.

Nobody-solved-this is not a claim we make. Mock ERC-20s stand in for the lent asset; the source event, the proof, the verification and the refusal are all real. Full limitation list and prior-art assessment in the repository.